

Souvik Roy

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SUMMARY

Results-driven Full Stack Developer with expertise in JavaScript, React, and Node.js, expanding into AI/ML engineering through hands-on work with Sentence Transformers, FAISS, and Groq LLMs. Continuously upskilling in agentic AI and applied machine learning through recent certifications, with a focus on building intelligent, scalable, real-world software.

SKILLS

Programming: Python, JavaScript, C++
Frontend: React.js, HTML5, CSS3, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, REST APIs
AI/ML: Sentence Transformers, FAISS, Groq LLM, Prompt Engineering, NLP
Database: MongoDB, SQL
Tools: Git, GitHub, Hugging Face, Railway, Vercel, Render

EXPERIENCE

Full Stack Web Developer Intern Nov 2025 - Dec 2025

CodeAlpha — Kolkata, West Bengal

- Developed responsive full stack web applications using MongoDB, Express, React, and Node.js.
- Debugged and resolved issues in various web applications to enhance performance.
- Tested websites across multiple browsers to ensure compatibility and responsiveness.
- Created technical documentation of all projects developed during the internship period.

AI/ML Intern (Project-Based Experiential Learning - PBEL) June 2026 - July 2026

IBM Developer Skills Network — Remote

- Developed CareerCompass-AI, an AI-powered career recommendation platform using FastAPI, React, Sentence Transformers, FAISS, and Groq LLMs for semantic resume analysis and personalized career guidance.
- Built a semantic retrieval engine that matched resumes against 140K+ job descriptions using Sentence Transformers + FAISS, enabling context-aware job recommendations.
- Optimized cloud deployment by reducing inference data from ~270 MB to ~16 MB, enabling reliable deployment on Railway while improving memory efficiency.
- Deployed the complete full-stack application using Railway (backend), Vercel (frontend), and Hugging Face (model and embedding artifacts).

PROJECTS

ET Story Arc Tracker | React, Express, Node, MongoDB and Groq | [GitHub](#) | [Live Demo](#) March 2026

- Designed and implemented a 3-agent AI pipeline that transformed live news into structured, personalized insights using Llama models and intelligent model routing.
- Engineered a full-stack React and Node.js application featuring REST APIs, CORS-secured requests, and multi-platform deployment.
- Designed and built a multi-turn chat system with conversation persistence, typing indicators, and hook-based state management.
- Integrated Tavily Search API with caching, fallback mechanisms, and robust error handling for reliable real-time news ingestion.

CareerCompass-AI | Python, FastAPI, React.js, Tailwind CSS, Sentence Transformers, FAISS | [GitHub](#) | [Live Demo](#) June 2026 - July 2026

- Developed an end-to-end AI career platform performing resume parsing, semantic job matching, skill-gap analysis, and personalized guidance using LLMs.
- Engineered a semantic retrieval system with Sentence Transformers + FAISS to match resumes against 140K+ job descriptions, enabling context-aware recommendations beyond keyword matching.
- Built and deployed a FastAPI backend and React.js frontend, integrating Groq LLMs for AI-generated career reports with personalized learning roadmaps.
- Optimized cloud inference by reducing deployment size from ~270 MB to ~16 MB, enabling stable, memory-efficient deployment on Railway, Vercel, and Hugging Face.

EDUCATION

Bachelor of Engineering - Civil Engineering Nov 2024 - Dec 2028

Jadavpur University — Kolkata, West Bengal

Higher Secondary Education - Science (PCM) May 2024

St. Patrick's Higher Secondary School — Asansol, West Bengal

CERTIFICATIONS

- Complete Agentic AI and MCP Course by Ed Donner - Udemy, Issued July 2026
- AI and Machine Learning, Data Science Bootcamp by Daniel Bourke - Udemy, Issued February 2026
- AI Tools and ChatGPT Workshop - be10X, Issued 2025